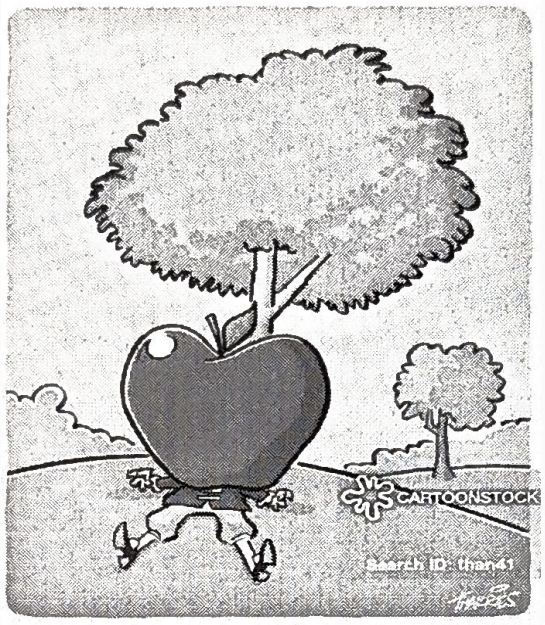
Learning Goals (Students will be able too…)

- Describe Newton's Three laws of motion in their own way.
- Solve problems involving Newton's three laws.



Newton regretted dabbling in genetic modification.

Newton's First Law of Motion

Object in motion will stay in motion, and an object at rest will stay at rest unless acted upon by an unbalanced force, in which it will change direction and speed.

- **Newton's First Law = Law of Inertia**
- **Implications of Newton's First Law:**
  - A net force is not required for an object to maintain a constant velocity.
  - A net force is required to change the velocity of an object in magnitude, direction, or both.
  - External forces are required to change the motion of an object. Internal forces have no effect on an object's motion.
- **Inertia:** a measure of an object's resistance to change motion.
  - Object at rest will stay at rest until a net force act on it.
  - Object in motion will maintain a constant velocity until a new force act on it.

Newton's Second Law of Motion

If the net external force on an object is not zero, the object will accelerate in the direction of the net force. The magnitude of the acceleration is directly proportional to the magnitude of the net force and inversely proportional to the object's mass. $\sum \bar{F} = m\bar{a}$ (Net force is the sum of all the individual external forces)

Basically, Newton's second law is saying that if you apply more force to an object its going to acceleration at a higher rate.

Newton's Third Law of Motion

For every action, there is a simultaneous equal and opposite reaction.

Weight & Normal Force

- **Weight:** the gravitational force of Earth on an Object ($\vec{F} = mg$)

Read: pg. 74-75 of the textbook regarding weight and its relation to force of **normal** as a review from grade 11.

Newton's Law Conceptual - Classroom Examples

1) Can you think of some real-life examples of when?
   a. Newton's First Law is at play?
   b. Newton's Second Law is at play?
   c. Newton's Third Law is at play?

a) This page is not moving on its own

b) Heavier vehicles accelerate slower (assuming equal net force)

c) If you push a wall, the wall will push back on you

Newton's First Law- Classroom Examples

1) **True or False:** Objects in orbit around Earth (like a satellite) must have a net force acting on them. Justify your answer.

True. Satellites are constantly experiencing the force of gravity, which keeps them in orbit.

2) Cristiano Ronaldo is racing against Neymar to get to the soccer ball that is just laying on the field. Cristiano gets there first and kicks the ball into the air.
   a. Why is this a case of Newton's first law?

The ball remains stationary until Ronaldo kicks it.

   b. Is this an example of something remaining at rest or remaining in motion?

Something remaining at rest.

3) Think about an example where **Newton's First law** plays a hand in **safety** within our everyday lives. What is happening in those scenarios that help **keep us safe**?

If yon get into a car crash, your body keeps moving forward until the seat belt can stop you safely.

Newton's Second Law- Classroom Examples

A falling object has only one force acting on it – gravity (ignoring air resistance)

1) What is the acceleration of a freely falling object?

$F_{net} = ma$ $\quad ma = -mg$

$F_{net} = -mg$ $\quad (a = -g)$ $\quad \boxed{a = -9.81\,m/s^2}$

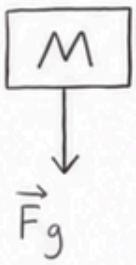
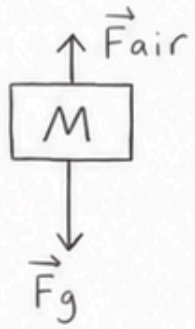
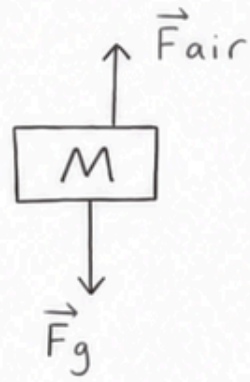
2) If I am standing straight up (not moving just standing), what are the forces acting upon me? What is my **NET** force then??

$F_{net} = ma$ | $F_{net} = F_N - F_g$ $\quad \therefore$ the net force is 0N.

$F_{net} = 0$ | $0 = F_N - F_g$

$F_N = F_g$

3) Recall that **air resistance** is proportional to **speed squared**. Imagine you drop an object off a tall building from rest. Draw a free body diagram of the object showing the forces of gravity and air resistance (it is important you draw the vectors of an appropriate length to correctly depict the situation)

When it first leaves your hand	A short time after it leaves your hand when it is still travelling slowly	A bit later when it is travelling more quickly
		
Explain The velocity when it first leaves your hand is 0, so there is no air resistance.	***Explain*** There is now a velocity, so air resistance comes into effect.	***Explain*** The velocity has incrensed, so air resistanie also increases.

4) Can the force of air resistance ever become larger than the force of gravity for a free-falling object? **Explain** (consider Newton's second law that discusses object that accelerate when there exits an unbalanced force).

No, for an object dropped from rest in unchanged conditions. Air resistance increases with velocity, but once it equals the force of gravity, velocity will remain constant, and so will air resistance.

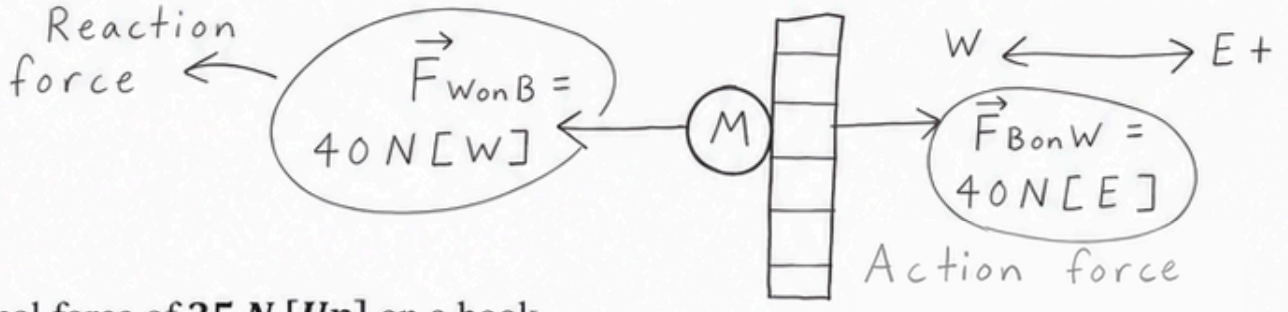
Newton's Third Law- Classroom Examples

Here is a video of some gravity defying tennis shots just for fun: https://www.youtube.com/watch?v=MGSr-WrdvWY

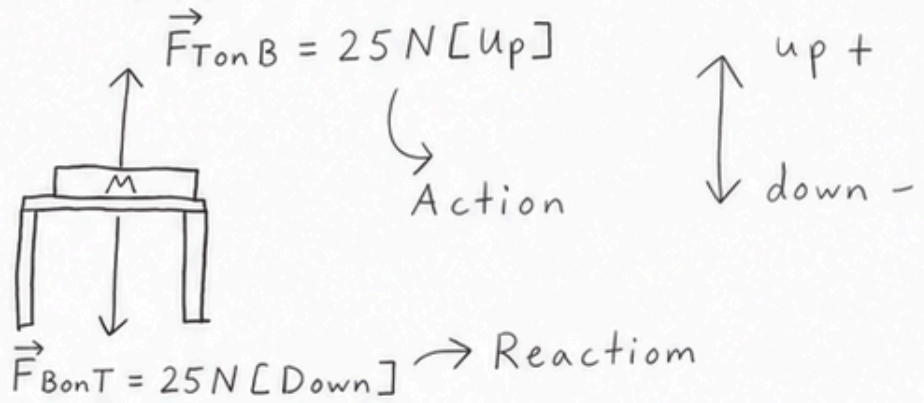
- Tennis a beautiful sport because we you can see **Newton's laws all** in action (and other reasons). During a serve, the mass of the tennis ball remains constant, but the more force you apply to the constant mass of the tennis ball onto the serve, the greater the acceleration will be. What law is this an example of? Law 2
- Whenever a tennis player strikes a ball the force of the racquets striking the ball and the force of the ball striking the racquet both **act** upon each other in equal but opposite ways. What law is this an example of? Law 3

1) For each of the following "**action forces**" state the **reaction force** (don't forget to include the direction). Draw a mini picture of the scenario.

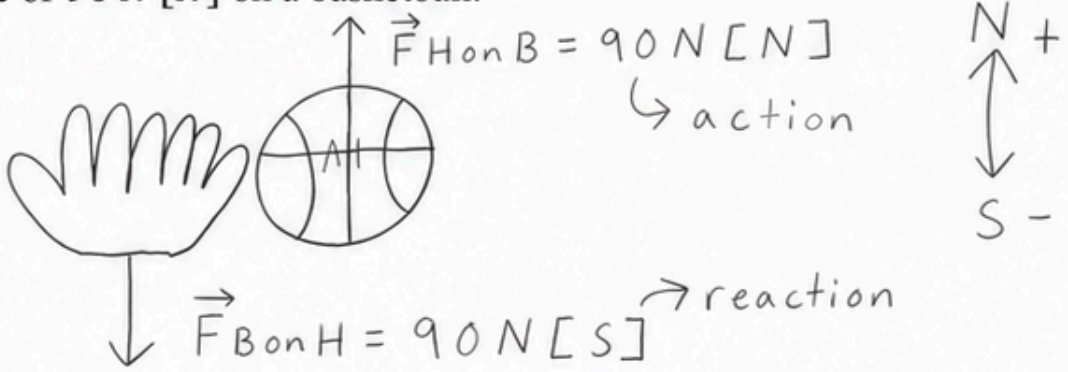
a. You are playing tennis with the wall. The ball exerts a force of **40*N* [*E*]** onto the wall.



b. A table exerts a normal force of **25 *N* [*Up*]** on a book.



c. Your hand exerts a force of **90 *N* [*N*]** on a basketball.



Homework

Pg 74 # 1, 2, 4

AND

Khan Academy : https://www.khanacademy.org/science/physics/forces-newtons-laws

- Do all **3 conceptual practice questions** (scroll under the list of videos- which you can use as a resource) and do "Newton's First Law", "Newton's Third Law of Motion" and " All of Newton's laws of Motion."